Cyber Power as a Structural Variable of Maritime Power

A conceptual update of its formulation

EXTENDED MARITIME POWER FORMULA

$$MP = \left[\left(MI \times PN \right) \times SW \times MS^{\gamma} \times CP^{\alpha} \right]^{1/n_{eff}}$$

$n_{eff} = 3+\gamma+\alpha$ · $MP_{norm} \in (0,1]$ · Normalization: geometric mean (UNDP 2010; OECD 2008)

ORIGINAL CONTRIBUTIONS OF THE ARTICLE

- **Cyber power (CP)** as a structural variable of maritime power — first formalization in the Spanish-language literature.
- **Strategic will (SW)** repositioned as a multiplier of the complete MP system, not only of naval power.
- **Maritime security (MS)** disaggregated into three dimensions: interstate conflicts, terrorism, and blue crime.
- **Normalization via n_{eff}** → $MP_{norm} \in (0,1]$ with a directly interpretable five-level scale.
- **CP is the most efficient investment vector** for middle powers ($\alpha < 1$): larger gap, high initial returns, transversal enabling character.

5	2	5	100k
structural variables formalized	disaggregation levels (macro + 27 sub-indicators)	levels of the analytical scale (0,1]	Monte Carlo verification trials

COLOMBIA — MP PROFILE *ref: Vargas Suárez et al., 2021*

0.53

★ MODERATE LEVEL

MI	NP	SW	MS	CP
0.60	0.55	0.48	0.59	0.40

COMMON-SCALE ANCHORING · INDEX 0.44 (RANK 33/99)

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Figure 1. Extended maritime power formula — two-level architecture

Level 1 architecture (general normalized expression) and Level 2 (disaggregation by component)

LEVEL 1 — GENERAL NORMALIZED EXPRESSION

$$MP_{norm} = [\ (IM \times PN) \times SW \times MS^{\gamma} \times CP^{\alpha}\ ]^{1/(n_{eff})}$$

$$n_{eff} = 3 + \gamma + \alpha \cdot MP_{norm} \in (0,\ 1] \cdot \text{All variables} \in (0,\ 1]$$

Preserves MP = MI × NP (Solís Oyarzún, 1997; Uribe Cáceres et al., 2016) · Normalization: geometric mean — UNDP (2010); OECD (2008) · Common-scale anchoring: Colombia → MP_{norm} = 0.53, same scale as Vargas Suárez et al. (2021, p. 295); the contribution is decomposing it

LEVEL 2 — COMPONENT DISAGGREGATION

MI

Maritime Interests

Political, economic, and security benefits the State obtains from the sea.

$MI = \Sigma\ w_i \cdot MI_i$

$\Sigma w_i = 1;\ w_i \in (0,1);\ MI \in (0,1]$

Weighted sub-index
Colombia: n = 18 national MI

SOURCES

Till (2007)
Uribe Cáceres et al. (2016)
CCO (2021); OECD (2008)

NP

Naval Power

Military capacity to use, protect, and control the maritime spaces.

$NP = F \times P \times MDA^{\mu}$

$NP \in (0,1];\ \mu = \text{MDA calibration}$

F: force+C2+tactical CD
MDA elevated to structural level

SOURCES

Mahan (1890); Corbett (1911); Castex (1931)
Till (2007); Bebber (2022)
Uribe Cáceres et al. (2016); Cabuya-Padilla (2024)

SW

Strategic Will

National determination to project and sustain the maritime effort.

$SW = (PW^{w_1} \cdot IW^{w_2} \cdot SocW^{w_3})^{1/\Sigma w}$

$SW \in (0,1];\ \Sigma w_i = 1$

Weighted geometric mean
MC transversal to PW·IW·SocW

SOURCES

Uribe Cáceres et al. (2016); Till (2007)
Alba-Rocha et al. (2022); Mora Peña & Uribe (2020)
Albarán Gómez (2019); UNDP (2010); Realpe Díaz (2024)

MS

Maritime Security

Degree of control over threats and risks in the maritime domain.

$MS = (IC^{a_1} \cdot TE^{a_2} \cdot BC^{a_3})^{1/\Sigma a}$

$MS \in (0,1];\ \Sigma a_i = 1;\ \gamma = \text{global MS weight}$

IC: interstate conflicts and gray zone
TE: terrorism · BC: blue crime

SOURCES

Bueger & Edmunds (2024)
IMO (2017; 2021)
Vargas Suárez et al. (2021)

CP

Cyber Power — central contribution of the article

Capacity to operate, defend, and project power in maritime cyberspace.

$CP = \min\{1,\ (\beta_1 \cdot CS_m + \beta_2 \cdot CD_m + \beta_3 \cdot CR_m) \cdot (1 + \delta \cdot CDip)\}$

$CP \in (0,1];\ \Sigma\beta_i = 1;\ \alpha(SCMI) < 1 \rightarrow \text{concave: initial returns} \uparrow$

CS_m: cyber security · CD_m: cyber defense
CR_m: cyber resilience · CDip: cyber diplomacy

SOURCES

Nye (2010); Bebber (2022); Cano (2024)
Cabuya-Padilla (2024, 2025)
Realpe Díaz (2024); Caro Bejarano (2013)

TRANSVERSAL METHODOLOGICAL DECISIONS

MDA: not an independent variable — distributed as an enabling factor across NP(F), MS, and CP (Bebber, 2022; Bueger & Edmunds, 2024)

Maritime Consciousness (Uribe Cáceres et al., 2016, p. 64): a cultural-identity construct, distinct from technical-operational MDA — transversal to PW·IW·SocW in SW

CDip (Realpe Díaz, 2024, pp. 297–299): dual role — projection coefficient in CP *and* expression of PW in SW, without analytical duplication. Model of an analytical-conceptual nature, not computational (Solís Oyarzún, 1997; Till, 2007)

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A conceptual update of its formulation — from a middle regional power's perspective

CF Diego Edison Cabuya Padilla, PhD · Naval Academy for Strategic Studies · Bogota, Colombia · 2026

THE PROBLEM

The **classical formula** of maritime power (Maritime Interests × Naval Power) does not capture the dynamics of **digitalized maritime environments**, where navigation, traffic, port facilities, and ocean surveillance depend on interconnected digital infrastructures vulnerable to cyberattacks.

FROM THE CLASSICAL TO THE EXTENDED FORMULA

$MP = MI \times PN$

CLASSICAL · 2 VARIABLES

→

$$MP_{norm} = [(MI \times NP) \times SW \times MS^{\gamma} \times CP^{\alpha}]^{1/n_{eff}}$$

EXTENDED · 5 VARIABLES

MI	NP	SW	MS	CP
Maritime Interests	Naval Power	Strategic Will	Maritime Security	Cyber Power

FIVE-LEVEL INTERPRETATION SCALE

Critical	Low	Moderate	Significant	High
0.00–0.20	0.21–0.40	0.41–0.60	0.61–0.80	0.81–1.00

THE EVIDENCE — COLOMBIA CASE STUDY

0.53	33/99	CP	100k
Normalized maritime power (Moderate)	Global maritime power ranking	Most efficient investment vector	Monte Carlo verification trials

FIVE ORIGINAL CONTRIBUTIONS

1. **Cyber power** formalized as a fifth structural variable of maritime power.
2. **Strategic will** repositioned as a multiplier of the complete system.
3. **Maritime security** disaggregated into three threat dimensions.
4. **Normalization** yielding an interpretable index in (0, 1].
5. **National Maritime Power Index (NMPI)** proposed as a comparative research agenda, with an open-access simulator.

Cyber Power as a Structural Variable of Maritime Power · Cabuya Padilla, D. E. (2026)
Naval Academy for Strategic Studies (ANEES) · Interactive simulator: github.com/diegocabuya/Cyber-Power-and-Maritime-Power